Probability as Constraint Geometry:

Toward a Derivation of the Born Rule from Closure Dynamics

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Abstract

Paper XXIX defined the observer as a constraint substructure within the VFD event-order framework and showed that measurement corresponds to constraint-induced reduction of admissible configurations. The present paper addresses the next gap: *why do admissible outcomes have different probabilities, and why do those probabilities take the quadratic form* $P_i = |\psi_i|^2$*?*

Admissibility alone determines which outcomes are possible; it does not determine their relative frequencies. This paper proposes that outcome probabilities arise as geometric weights over the set of observer-admissible configurations, with weights determined by the structure of the closure dynamics rather than by primitive stochastic postulates.

Four candidate weighting routes are examined: basin-volume measure under the crystallisation flow, stability-depth weighting from the closure functional, stationary-measure inheritance from the Paper XIV equilibrium distribution $P_{\mathrm{st}} \propto e^{-2\mathcal{F}/\sigma^2}$, and a Gleason-type argument showing that quadratic weighting is the unique compatible measure once the admissible-sector geometry has Hilbert-space structure. The stationary-measure route is the strongest: in the equilibrium regime identified in Paper XIV, where the Nelson density satisfies $|\Psi|^2 = \rho = P_{\mathrm{st}}$, the effective Born measure coincides with the stationary measure of the closure dynamics. A two-basin toy model demonstrates the mechanism explicitly.

The paper does not claim a complete derivation of the Born rule. It identifies a candidate substrate explanation: probability as the induced measure over observer-compatible admissible sectors, with quadratic weighting emerging from the equilibrium structure of the closure dynamics under suitable assumptions.

1 Introduction

The Born rule — the prescription that the probability of measuring outcome i is $P_i = |\langle\phi_i|\psi\rangle|^2$ — is one of the most empirically successful postulates in physics. It is also one of the least explained. Standard quantum mechanics introduces it as an axiom, without providing a mechanism for why measurement outcomes should be weighted by the squared modulus of the amplitude.

The gap

The VFD programme has addressed several foundational questions in sequence:

- Papers XII–XIV: closure dynamics, stochastic extension, stationary distribution.
- Papers XV–XXI: quantum structure recovery (Schrödinger as equilibrium tangent limit).
- Paper XXII: spectral bridge connecting mass, gauge, and constants.
- Papers XXIII–XXVII: gravitational structure from event-order geometry.
- Paper XXVIII: unification bridge between quantum and gravitational tracks.
- Paper XXIX: observer placement as constraint substructure.

Paper XXIX showed that measurement corresponds to constraint-induced reduction of admissible configurations. But admissibility is binary: a configuration is either admissible or not. This does not explain why different admissible outcomes are realised with different frequencies. The Born-rule gap remains:

> Given a set of observer-admissible outcomes, what determines their relative probabilities?

Proposal

This paper proposes that outcome probabilities are not primitive stochastic ingredients added to measurement. They are the induced measure over observer-compatible admissible configurations, with weights determined by the geometry and dynamics of constraint satisfaction.

Goals

This paper has two goals:

1. Explain why admissibility must be supplemented by a weighting measure to account for observed frequencies.
2. Identify the equilibrium stationary measure of the closure dynamics as the natural candidate for that weighting.

Scope

The paper develops four candidate routes toward quadratic weighting, works a toy model explicitly, and identifies the conditions under which the Born rule emerges. It does not claim a complete, assumption-free derivation. It provides a candidate substrate explanation and identifies what remains open.

2 Programme Position

The logical chain from prior papers:

1. **Structure.** The 600-cell provides the geometric substrate (Papers I–V, XXII).
2. **Dynamics.** Closure gradient flow and stochastic extension (Papers XII–XIV).
3. **Quantum recovery.** Schrödinger equation as equilibrium tangent limit (Papers XV–XXI).
4. **Observer.** Bounded, stable, self-referential constraint substructure (Paper XXIX).
5. **Measurement.** Reduction to observer-admissible configurations (Paper XXIX).
6. **Probability.** Geometric weighting over admissible sectors (*this paper*).

Step 6 is the natural continuation: once measurement is understood as admissibility reduction, the remaining task is to explain the weighting.

3 Formal Setup

3.1 Configuration space and admissibility

Let Φ denote the space of field configurations on the event set $\mathcal{E}$. The closure functional $\mathcal{F} : \Phi \to \mathbb{R}_{\geq 0}$ assigns a closure cost to each configuration.

Definition 1 (Observer-admissible configurations)**.** *Given an observer* $(\mathcal{O}, \mathcal{C}_\mathcal{O})$ *as defined in Paper XXIX, the admissible set is:*

$$\Phi_{\text{adm}}(\mathcal{O}) = \left\{ \Phi \in \Phi \;\middle|\; \mathcal{F}[\Phi] \textit{ locally minimal}, \; \mathcal{C}_\mathcal{O}\big(\Phi|_\mathcal{O}, \partial_\mathcal{O}\Phi\big) \leq \epsilon \right\} \tag{1}$$

3.2 Outcome sectors

When the admissible set contains multiple locally stable configurations, we partition it into *outcome sectors*:

$$\Phi_{\text{adm}}(\mathcal{O}) = \Phi_1 \cup \Phi_2 \cup \cdots \cup \Phi_N \tag{2}$$

where each Φ_i is a connected component of the admissible set (a basin neighbourhood of a distinct local minimum of $\mathcal{F}$).

3.3 The probability question

Admissibility determines which sectors are possible. It does not determine their relative likelihood:

$$\Phi_i \in \Phi_{\text{adm}}(\mathcal{O}) \;\text{ for all } i \quad \not\Rightarrow \quad P(\Phi_i \mid \mathcal{O}) = \tfrac{1}{N} \;\text{ for all } i \tag{3}$$

The task is to construct a probability measure over outcome sectors from the structure of the closure dynamics.

4 Why Admissibility Alone Is Not Enough

Consider a system with two admissible outcome sectors Φ_1 and Φ_2. Both satisfy the observer constraint. Both are locally stable under the closure functional. If admissibility were the whole story, the prediction would be $P_1 = P_2 = 1/2$ for any two-outcome measurement.

This contradicts experiment. Quantum mechanics predicts — and experiment confirms — that outcome probabilities depend on the state: $P_i = |\langle\phi_i|\psi\rangle|^2$, which is generically not $1/N$.

The gap is structural:

> Admissibility is a binary filter. Probability requires a measure. The filter determines the support; the measure determines the weight.

What is needed is a principled construction of a weight w_i for each admissible sector, such that:

$$P(\Phi_i \mid \mathcal{O}) = \frac{w_i(\mathcal{O})}{\sum_{j=1}^{N} w_j(\mathcal{O})} \tag{4}$$

The remainder of this paper develops four candidate routes to w_i. Routes A and B motivate why probability should be geometric in character. Route C provides the main dynamical route: the stationary measure of the closure dynamics. Route D provides structural uniqueness via Gleason's theorem. Routes C and D carry the paper's central argument; Routes A and B are preparatory.

5 Route A: Basin-Volume Weighting

Definition 2 (Basin of attraction)**.** *For a locally stable configuration Φ_i^* (local minimum of $\mathcal{F}$ in Φ_{adm}), the basin of attraction is:*

$$B_i = \left\{ \Phi \in \Phi \;\middle|\; \textit{gradient flow of } \mathcal{F} \textit{ starting at } \Phi \textit{ converges to } \Phi_i^* \right\} \tag{5}$$

Definition 3 (Basin-volume weight)**.**

$$w_i^{(\mathrm{A})} = \mu(B_i \cap \Phi_{\text{adm}}) \tag{6}$$

where μ is a natural measure on configuration space (e.g. Lebesgue measure on the finite-dimensional event set).

Interpretation. If the system's initial configuration is drawn from a distribution that is approximately uniform over the admissible region, then the probability of ending in sector i is proportional to the volume of its basin. Larger basins attract more initial conditions.

Remark 4 (Strengths and limitations)**.** *Basin-volume weighting is natural for classical dynamical systems. It explains why different outcomes have different probabilities without invoking randomness: the asymmetry comes from geometry. However, it does not by itself explain why the weights should be quadratic in any amplitude. It produces a probability measure, but not necessarily the Born measure.*

6 Route B: Stability-Depth Weighting

Definition 5 (Stability-depth weight)**.**

$$w_i^{(\mathrm{B})} = e^{-\beta\,\mathcal{F}[\Phi_i^*]} \tag{7}$$

where $\mathcal{F}[\Phi_i^*]$ is the closure-functional value at the i-th minimum and $\beta > 0$ is a scale parameter.

Interpretation. Configurations with lower closure cost (deeper minima) are more heavily weighted. This is the Boltzmann-like weighting familiar from statistical mechanics: outcomes that are more constraint-compatible are more probable.

Remark 6. *This route connects to the thermodynamic structure of the closure landscape. It produces non-equal probabilities whenever the minima have different depths. But the Boltzmann form $e^{-\beta E}$ is not the same as the Born form $|\psi|^2$ unless β and $\mathcal{F}$ are related to amplitude in a specific way. Route C makes this connection explicit.*

7 Route C: Stationary-Measure Inheritance

This is the strongest route, because it connects directly to established results from the programme.

7.1 The Paper XIV stationary distribution

Paper XIV [Smart, 2026a] established that the stochastic closure dynamics (Langevin equation $d\psi = -\nabla\mathcal{F}\,dt + \sigma\,dW$) admits a unique stationary distribution:

$$P_{\mathrm{st}}(\psi) \propto e^{-2\mathcal{F}(\psi)/\sigma^2} \tag{8}$$

This distribution concentrates on the closure manifold $\mathcal{M}_{\mathrm{cl}}$ and provides the probabilistic structure of the closure landscape.

7.2 The Nelson wavefunction connection

Paper XVIII [Smart, 2026c] constructed the Nelson wavefunction $\Psi = \sqrt{\rho}\,e^{iS/\sigma^2}$ from the paired forward/backward closure processes. At the Paper XIV equilibrium, the density ρ equals P_{st}, so:

$$|\Psi|^2 = \rho = P_{\mathrm{st}} \propto e^{-2\mathcal{F}/\sigma^2} \tag{9}$$

This is the central observation: *in the equilibrium regime, the effective Born density $|\Psi|^2$ coincides with the stationary measure of the closure dynamics.*

7.3 Why measurement samples the stationary sector measure

The connection between the stationary measure and measurement outcomes rests on the following reasoning:

1. The underlying stochastic closure dynamics has a unique stationary measure P_{st} (Paper XIV).

2. Observer constraints carve this measure into admissible sectors (Paper XXIX).

3. Measurement frequencies correspond to repeated sampling from the constrained stationary ensemble: each measurement draws from the same P_{st}, restricted to the admissible set.

This paper treats this reasoning as a working hypothesis rather than a completed theorem. In particular, the assumption that the system has equilibrated to P_{st} at the time of measurement is not proved for general measurement contexts.

7.4 Stationary-measure weight

Definition 7 (Stationary-measure weight)**.**

$$w_i^{(\mathrm{C})} = \int_{\Phi_i} P_{\mathrm{st}}(\Phi)\, d\mu(\Phi) = \int_{\Phi_i} |\Psi(\Phi)|^2\, d\mu(\Phi) \tag{10}$$

the total stationary probability mass in outcome sector Φ_i.

Proposition 8 (Conditional Born weighting from stationary sector measure)**.** *Under the following assumptions:*

1. *The system is at or near the Paper XIV equilibrium ($\rho \approx P_{\mathrm{st}}$).*

2. *The outcome sectors Φ_i are well-separated basins (negligible overlap).*

3. *The Nelson wavefunction Ψ admits a decomposition $\Psi = \sum_i c_i \phi_i$ where ϕ_i is supported on Φ_i.*

Then:

$$P(\Phi_i \mid \mathcal{O}) = \frac{w_i^{(\mathrm{C})}}{\sum_j w_j^{(\mathrm{C})}} = \frac{\int_{\Phi_i} |\Psi|^2\, d\mu}{\int_{\Phi_{\mathrm{adm}}} |\Psi|^2\, d\mu} = \frac{|c_i|^2 \|\phi_i\|^2}{\sum_j |c_j|^2 \|\phi_j\|^2} \tag{11}$$

If the ϕ_i are normalised ($\|\phi_i\| = 1$), this reduces to:

$$P(\Phi_i \mid \mathcal{O}) = |c_i|^2 \tag{12}$$

Proof. By assumption 3, $|\Psi|^2 = |\sum_i c_i \phi_i|^2$. By assumption 2, the supports of the ϕ_i are effectively disjoint up to negligible overlap on the relevant integration domains, so cross terms $c_i^* c_j \phi_i^* \phi_j$ are negligible when integrated over any single sector Φ_i. Therefore $\int_{\Phi_i} |\Psi|^2 = |c_i|^2 \|\phi_i\|^2$. The result follows by normalisation. □

Remark 9 (What this does and does not establish)**.** *Proposition 8 shows that if the system is near equilibrium, if outcome sectors are well-separated, and if the wavefunction decomposes into sector-supported components, then the Born rule follows from the stationary measure of the closure dynamics. The assumptions are not trivial: they require the system to be in the regime where the Schrödinger approximation holds (Papers XVIII–XXI). Whether these assumptions are satisfied in all measurement contexts is an open question.*

Remark 10 (Logical status)**.** *Within the equilibrium closure regime, the quadratic probability law need not be introduced as an independent measurement axiom; it is inherited from the stationary measure. The chain is:*

1. *the stationary distribution* $P_{\mathrm{st}} \propto e^{-2\mathcal{F}/\sigma^2}$ *(derived in Paper XIV),*

2. *the identification* $|\Psi|^2 = P_{\mathrm{st}}$ *at equilibrium (derived in Paper XVIII),*

3. *the decomposition of* P_{st} *into outcome-sector integrals (Definition 7).*

The quadratic form is not imposed; it is inherited from the equilibrium structure. Whether this inheritance extends to all measurement contexts, including those far from equilibrium, is an open question.

Remark 11 (Three levels of identification)**.** *Three levels should be distinguished. First,* P_{st} *is the exact stationary measure of the stochastic closure process (Paper XIV). Second, in the equilibrium tangent regime of Papers XVIII–XXI, this stationary density is represented effectively as* $|\Psi|^2$*. Third, measurement frequencies are hypothesised to sample the induced sector masses of that equilibrium measure. The present paper argues that, under these identifications and assumptions, the observed quadratic law is inherited rather than postulated.*

8 Route D: Gleason-Type Argument

If the admissible-sector geometry has Hilbert-space structure, an independent argument constrains the form of the probability measure.

8.1 The Gleason bridge

Gleason's theorem [Gleason, 1957] states: on a Hilbert space $\mathcal{H}$ of dimension ≥ 3, the only non-contextual, countably additive probability measure μ on the lattice of closed subspaces satisfies $\mu(P) = \mathrm{tr}(\rho P)$ for some density operator ρ. For a pure state $\rho = |\psi\rangle\langle\psi|$, this gives:

$$P_i = \mathrm{tr}(|\psi\rangle\langle\psi| \cdot \Pi_i) = |\langle\phi_i|\psi\rangle|^2 \tag{13}$$

the Born rule.

8.2 Application to the VFD framework

Papers XVIII–XXI established that the equilibrium tangent dynamics of the closure framework is the Schrödinger equation, which naturally lives on a Hilbert space. If the observer-admissible outcome sectors correspond to orthogonal subspaces of this effective Hilbert space, and if the probability measure over sectors is required to be non-contextual and additive, then Gleason's theorem implies the Born rule.

Remark 12 (What VFD adds to Gleason)**.** *Gleason's theorem assumes Hilbert-space structure and derives the Born rule. Route D does not provide the physics; it provides a uniqueness confirmation. The VFD contribution is to explain why the relevant space has Hilbert structure: because the equilibrium tangent dynamics is Schrödinger-type (Papers XVIII–XXI). The Gleason argument then confirms that, given this structure, the Born rule is the only compatible measure. The physical origin of the weighting comes from Route C (the stationary measure); Route D establishes that no other weighting is consistent with the emergent Hilbert structure.*

Remark 13 (Assumptions)**.** *This route requires: (1) the system is in the equilibrium tangent regime; (2) the outcome sectors correspond to orthogonal subspaces; (3) the dimension is ≥ 3; (4) the probability measure is non-contextual and additive. These are the standard Gleason assumptions. Whether the VFD framework satisfies them in full generality is an open question.*

9 Toy Model: Two-Basin Weighting

This is a schematic effective model illustrating sector weighting. It is not derived from the full 600-cell construction or from the double-well closure potential alone; it demonstrates the mechanism by which the stationary measure produces Born-like weights when an asymmetric wavefunction is decomposed into well-separated sectors.

Example 14 (Two-basin model)**.** *Consider a one-dimensional closure functional with two minima:*

$$\mathcal{F}(\psi) = (\psi^2 - 1)^2 \tag{14}$$

This has minima at $\psi^* = \pm 1$ with $\mathcal{F}(\pm 1) = 0$, and a maximum at $\psi = 0$ with $\mathcal{F}(0) = 1$.

Symmetric case. *The stationary distribution is $P_{\text{st}}(\psi) \propto e^{-2(\psi^2-1)^2/\sigma^2}$. By symmetry, $\int_{-\infty}^{0} P_{\text{st}} = \int_{0}^{\infty} P_{\text{st}} = 1/2$. The two basins have equal weight: $P_1 = P_2 = 1/2$.*

Asymmetric case. *Now consider a state displaced from symmetry. Define the Nelson wavefunction:*

$$\Psi(\psi) = a\,\phi_1(\psi) + b\,\phi_2(\psi) \tag{15}$$

where ϕ_1 is a normalised Gaussian localised near $\psi = -1$ and ϕ_2 is localised near $\psi = +1$, with $|a|^2 + |b|^2 = 1$.

The stationary-measure weights are:

$$w_1 = \int_{-\infty}^{0} |\Psi|^2\, d\psi \approx |a|^2 \quad \textit{(cross terms negligible by separation)} \tag{16}$$

$$w_2 = \int_{0}^{\infty} |\Psi|^2\, d\psi \approx |b|^2 \tag{17}$$

Therefore:

$$P_1 = |a|^2, \qquad P_2 = |b|^2 \tag{18}$$

This is the Born rule for the two-outcome case.

Mechanism. *The asymmetric weighting arises not from external randomness but from the distribution of the stationary measure across basins. A state with more amplitude in basin 1 has more stationary probability mass in basin 1. The Born rule is the statement that the stationary measure decomposes according to squared amplitudes.*

10 Relationship to Standard Quantum Theory

10.1 What is preserved

Standard quantum predictions remain unchanged at the operational level. The VFD proposal does not modify the Born rule; it seeks to explain why it has the form it does. Amplitudes remain the effective computational tool. The proposal is a *substrate explanation of measurement weighting*: it identifies a dynamical mechanism (the stationary

measure of closure dynamics) that produces the same probability assignments as the Born postulate.

10.2 Comparison with other approaches

Table 1: Born-rule status across interpretive frameworks.

Framework	Born rule status	Mechanism
Copenhagen	Postulated axiom	None (primitive)
Many-worlds	Derived (decision/symmetry)	Branch counting / envariance
Bohmian mechanics	Derived (quantum equilibrium)	Typicality of initial conditions
Objective collapse	Modified / stochastic	Added stochastic law
VFD (this paper)	Candidate emergence	Stationary measure of closure dynamics

Remark 15. *The VFD route is closest in spirit to the Bohmian quantum-equilibrium approach: both derive the Born distribution from an underlying dynamical measure. The key difference is that the Bohmian approach assumes pilot-wave dynamics and derives* $|\psi|^2$ *as the equivariant measure, while VFD derives* $|\Psi|^2$ *as the stationary measure of the closure dynamics. The Bohmian approach requires a separate justification for why the universe is in quantum equilibrium; the VFD approach proposes that equilibrium is the generic long-time behaviour of the stochastic closure process.*

11 Route Comparison and Synthesis

Table 2: Comparison of the four weighting routes.

Route	Weight w_i	Born-like?	Status
A: Basin volume	$\mu(B_i \cap \Phi_{\text{adm}})$	Not necessarily	Natural but incomplete
B: Stability depth	$e^{-\beta\mathcal{F}[\Phi_i^*]}$	Boltzmann, not Born	Thermodynamic analogue
C: Stationary measure	$\int_{\Phi_i} \lvert\Psi\rvert^2 \, d\mu$	Yes (at equilibrium)	Strongest route
D: Gleason bridge	$\lvert\langle\phi_i\lvert\psi\rangle\rvert^2$	Yes (by theorem)	Requires Hilbert structure

Routes C and D are complementary:

- Route C provides the *dynamical* origin: in the equilibrium regime, the stationary measure of the closure process is represented effectively by $|\Psi|^2$.
- Route D provides the *structural* uniqueness: once Hilbert structure is present, the Born rule is the only compatible measure.

Together, they suggest: the closure dynamics produces an effective Hilbert-space structure at equilibrium (Papers XVIII–XXI), and the Born rule is recovered from the stationary sector measure of the dynamics (Route C) while also being the unique compatible measure on that structure (Route D).

Routes A and B are weaker individually but illuminate why probability is geometric: it arises from the structure of basins and the depth of constraint satisfaction, not from primitive randomness.

12 Discussion and Limitations

12.1 What is established

- Admissibility alone does not determine outcome probabilities; a weighting measure is required (Section 4).
- Four candidate weighting routes are examined, with explicit constructions for each.
- The stationary-measure route (Route C) yields Born-like weighting $P_i = |c_i|^2$ under stated assumptions (Proposition 8).
- The mechanism is demonstrated in a two-basin toy model (Example 14).
- The Gleason bridge (Route D) provides structural uniqueness: Born weighting is the only non-contextual additive measure on Hilbert space.
- Routes C and D are complementary: dynamical origin plus structural uniqueness.

12.2 What is not established

- A complete, assumption-free derivation of the Born rule. The result depends on the equilibrium assumption, well-separated basins, and sector-supported decomposition.
- Proof that the assumptions of Proposition 8 hold in all measurement contexts.
- Extension to continuous spectra or infinite-dimensional configuration spaces.
- A concrete construction of the weight measure from 600-cell geometry (the construction is abstract).
- Proof that the system reaches equilibrium in finite time for physically relevant configurations.
- Treatment of decoherence and environmental interaction within the constraint framework.
- The relationship between the coherence threshold ϵ and the basin structure.
- Proof that measurement interactions generically produce the well-separated sector structure assumed in Proposition 8. The "well-separated basins" assumption is doing significant work.
- Proof that the effective Hilbert-space decomposition $\Psi = \sum_i c_i \phi_i$ used in Proposition 8 is canonically determined by the observer-induced sector structure.
- Corrections to the Born rule away from equilibrium (the nonlinear residual of Paper XIX [Smart, 2026b] modifies the dynamics; its effect on probabilities is not analysed).

12.3 The equilibrium assumption

The strongest route (C) depends on the system being at or near the Paper XIV equilibrium. This is analogous to the quantum-equilibrium hypothesis in Bohmian mechanics: both frameworks derive the Born distribution from a dynamical measure, and both require a justification for why the system is in the relevant equilibrium state. Within VFD, the justification is that the stochastic closure dynamics has P_{st} as its unique stationary distribution; any initial distribution converges to P_{st} under ergodic evolution. Whether this convergence occurs on physically relevant timescales is an open question.

13 Future Directions

1. **Concrete constraint functional.** Constructing $\mathcal{C}_{\mathcal{O}}$ and the weight measure explicitly from the 600-cell geometry would move the framework from abstract construction to computable prediction.

2. **Away-from-equilibrium corrections.** The nonlinear residual δU_{rel} (Paper XIX) modifies the dynamics near but not at equilibrium. Its effect on outcome probabilities may produce measurable deviations from the Born rule — a potential experimental signature.

3. **Emergence of observer structures.** Showing how observer regions satisfying the Paper XXIX definition arise naturally from the closure dynamics would close the last gap between geometry and measurement.

4. **Decoherence integration.** The constraint framework should be compatible with decoherence as a mechanism for producing well-separated basins. Making this explicit would strengthen the assumptions of Proposition 8.

14 Conclusion

In this framework, probability is not a primitive stochastic ingredient added to measurement. It is the induced measure over the set of observer-compatible admissible configurations, with weights determined by the geometry and dynamics of constraint satisfaction.

The programme chain is now:

1. **Structure**: closure geometry on the 600-cell.

2. **Dynamics**: stochastic closure process with stationary distribution $P_{\mathrm{st}} \propto e^{-2\mathcal{F}/\sigma^2}$.

3. **Quantum recovery**: Schrödinger equation as equilibrium tangent limit, with $|\Psi|^2 = P_{\mathrm{st}}$ in the equilibrium regime.

4. **Observer**: constraint substructure determining admissible configurations.

5. **Measurement**: reduction to observer-admissible sectors.

6. **Probability**: in the equilibrium closure regime, stationary measure inherited by admissible sectors yields $P_i = |c_i|^2$.

In the equilibrium closure regime, the Born rule need not be introduced as an independent axiom. The Born weighting coincides with the stationary sector measure of the underlying closure process, and within the emergent Hilbert-space description it is the unique compatible measure under the usual Gleason assumptions. The assumptions under which this emerges — equilibrium, well-separated sectors, sector-supported decomposition — are stated explicitly, and what remains open is identified.

The standard quantum predictions are not modified. What changes is the explanatory status of the Born rule: from primitive postulate to candidate consequence of constraint geometry. A complete, assumption-free derivation for all measurement contexts remains an open target.

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